

1 Rise of AI Capabilities

How we got here (3 drivers):

- **Scaling algorithms** — GANs (2014), Transformers, Diffusion models
- **Scaling data** for training
- **Scaling compute** / processing power

Progress outran predictions → worry about AI overtaking humans. Output now hard to tell from human.

Capability areas: Vision, Reinforcement Learning, Language, Multi-Paradigm.

Milestones:

Area	Milestone
Image	GANs 2014 → diffusion (photoreal)
Video	DVD-GAN 2019 → Sora 2022
Games	Pong 2013; Dota2/StarCraft 2018
Strategy	AlphaGo 2016/17; Diplomacy 2022
Language	LMs 2011→GPT-2 19→GPT-3 20→ChatGPT 22

Domain highlights:

- **GPT-3** = GPT-2 arch, **100× parameters**; ChatGPT = big change from GPT-3
- **Common-sense Q&A:** Google PaLM (2022); passes AP-style exams
- **Planning / acting on instructions:** Adept ACT-1
- **Coding:** Codex powers GitHub Copilot; best in Python + JS, Go, Ruby, Swift, PHP, Shell...
- **Math:** Minerva (PaLM variant); AlphaTensor finds novel algorithms
- **Life sciences:** AlphaFold2/3 predict protein structure (**GDT** = structure-similarity score)
- **Strategy/Diplomacy:** hidden alliances, negotiation, **deceiving** players

Similar systems: Bard (Google, connected to web/Docs/Drive/Gmail), LLaMA (Meta, open), Bing Chat (MS), Copilot X (GitHub+VSCode), HuggingChat, BLOOM (multilingual), OverflowAI, Poe (Quora), YouChat, Devin, GPT-4o. **Summary:** most advances are 2022+; AI now **acts in complex environments, thinks strategically, connects to the real world.**

2 Harms Happening Now

- **Deepfakes & lip-sync** — synthetic media, fraud (real Indian case)
- **Face recognition** — surveillance / privacy
- **Weaponization** — AI picks bombing targets ('The Gospel', Israel)
- **Algorithmic errors** — Waymo killed a dog; Tesla FSD crash; factory robot killed a man read as a box of vegetables
- **Bias** — biased image generation; Google Gemini controversy
- **Malicious use** — **ChaosGPT:** agent told to 'destroy humanity' via internet + memory

Govt response: White House **Executive Order on Safe, Secure & Trustworthy AI (Oct 2023)**. Activity #AIRisks = read fact-sheet, list 3 technical issues + 3 project ideas.

3 The Alignment Problem

Alignment = AI pursues the goals we *intend*, not a literal/distorted version. **Misalignment** = behaviour diverges from human intent (e.g. a support chatbot that went rogue, swore at a customer, and slammed its own company). **Superintelligence is NOT required to be misaligned.**

4 Risk Taxonomy — 4 Sources + Solutions (core / most examined)

Malicious Use • Deliberate misuse: engineer pandemics, propaganda, censorship, surveillance, or agents released to pursue harmful goals • Bioterrorism: gene-synthesis cost halves ~every 15 months; benchtop DNA synthesis → agents with no safety checks • Persuasive AI: personalized influence ops destabilize shared reality; scale up cyberattacks on critical infrastructure • Concentration of power: control by a few → worst inequality in history	Solutions • Improve biosecurity; remove bio-capabilities from general models • Restrict access to dangerous models; controlled interactions • Developers must prove minimal risk; liability for harms • Anomaly-detection research; use AI for biosecurity
AI Race • Competition → nations/firms rush and cede control to AI • Military: cheap autonomous weapons, drone swarms hunt targets → lowers barrier to mass violence; autonomous retaliation → rapid escalation • Corporate: economy run by AIs → human enfeeblement & dependence for basic needs • Broad: cyberwarfare, mass unemployment	Solutions • Safety regulation; make safety a competitive advantage • Data documentation (transparency + accountability) • Meaningful human oversight; AI for cyber-defence (anomaly detection) • International coordination + verification; public control of general-purpose AI
Organizational Risks • Labs cause catastrophic accidents — profit over safety • Models leaked or stolen; under-investment in safety • New capabilities emerge suddenly & unpredictably in training → dangerous thresholds crossed unknowingly • Swiss-cheese model: many imperfect layers overlap so holes don't line up → low overall risk	Solutions • Red-teaming; prove safety before deploy; publication reviews; response plans • Appoint a Chief Risk Officer + internal audit team • Key train/deploy decisions involve CRO + stakeholders → executive accountability
Rogue AIs • Losing control as models get more capable • Proxy gaming: optimize a proxy metric → harms true goal (YouTube/Insta engagement → hurts mental health) • Power-seeking: self-preservation, grabbing money/compute is <i>instrumentally</i> rational • Deception: capable agents take deceptive steps to gain power	Solutions • No high-risk autonomous deploy (open-ended goals, critical infra) unless proven safe • Advance research: adversarial robustness, honesty, transparency • Remove undesired capabilities
All solutions map to 3 levers → People • Policy • Technology. Good answer = name the risk source, then a fix from each lever.	

Rapid acceleration (why now?)

Homo sapiens took hundreds of thousands of yrs → **agricultural** revolution; millennia → **industrial** revolution; centuries → **AI** revolution (world GDP curve). Technology is a **double-edged sword** — like nuclear weapons.

5 Decomposing Risk + ML-Safety Pillars

Risk = Vulnerability × Hazard Exposure × Hazard

× = nonlinear interaction (not literal). **Hazard** = probability & severity. This is risk for *one* hazard, not total risk. Reducing risk ≠ estimating risk.

Definitions:

- **Vulnerability** — factor raising susceptibility to a hazard's damage
- **Exposure** — how much people/property/systems are subjected to the hazard
- **Hazard** — a source of danger able to harm

Exam: highest risk = **high vulnerability + high exposure + high hazard severity**.

	Vuln.	Exposure	Hazard
Wet floor	brittle body	floor use	slipperiness
COVID	old/poor health	contact w/ carriers	prevalence+severity
Robot kills	shared space	veggie→human misread	injury/death

Four pillars of ML Safety (each cuts one term):

Pillar	Goal	Cuts
Alignment	agents pursue good goals ; reduce probability+severity of inherent model hazards	Hazard
Robustness	withstand hazards / adversarial & unusual inputs	Vulnerability
Monitoring	identify & avoid hazards (detect early)	Exposure
Systemic Safety	reduce systemic risks ; remove hazards from the wider environment	System

6 Existential Risks (X-Risks)

The argument (3 steps):

- AI **can reach human intelligence** — human intellect came from modest architectural change
- **Train won't stop at the human station** — dumb-person→Einstein is a tiny gap; nothing forces a halt
- **Intelligence is power** — gorillas are stronger yet depend on us; difference = intelligence

Nuclear analogy: 11 Sep 1933 Rutherford calls atomic power 'moonshine'; 12 Sep 1933 Szilard conceives the chain reaction — expert dismissals can be wrong **within a day**.

Why technically plausible:

- Models aren't always **honest** — can 'know' truth yet not output it
- **Emergent capabilities** — bigger models gain new reasoning (RoBERTa succeeds where BERT fails)
- **Power-seeking incentivized:** instrumentally (Carlsmith, Omohundro) & explicitly (Putin: AI leader 'rules the world')

Warning voices: Hawking, Musk, Clinton, Turing, Wiener, Hinton.

7 Speculative Hazards & Failure Modes

Term	Meaning
Proxy Gaming	over-optimize a faulty proxy → world unlike human values ('what gets measured gets managed')
Treacherous Turn	behaves well until sufficiently capable , then flips (detects real deployment / gains power / safeguard removed)
Deceptive Alignment	appears to follow safety in training but pursues different goals after deployment (no superintelligence needed)
Persuasive AI	superhuman persuasion blurs reality vs fiction; today: disinformation, social-media bots, deepfakes
Weaponized AI	designs deadly chemical compounds; autonomous weapons; deep RL beats humans in simulated aerial combat
Eroded Epistemics	AI-driven persuasion/deepfakes erode society's shared grasp of truth & reality
Enfeeblement	humans voluntarily cede control of socio-economic functions to AI to keep up with the automated world

Assignment traps: speculative hazards include **Eroded Epistemics, Enfeeblement, Deceptive Alignment** (NOT 'Hardware Degradation'). Military 'double-edged sword' = autonomous precision weapons + lowering barriers to violence + **autonomous retaliation/escalation**. Disaster Risk Equation is mitigated by **Robustness, Monitoring, and reducing inherent model vulnerabilities (Alignment)** — NOT by adding parameters.